Embedded System

Level: Bachelor

Programme: BE Computer

Semester: III/V

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Outline

Course Contents

1.Introduction to Embedded system (3 hrs)

2.Programming for Embedded systems (5 hrs)

3.Real-time operating systems (RTOS) (5 hrs)

4.Embedded System Design using VHDL (5 hrs)

5.Communications Protocals (3 hrs)

6.Pheripherals and Interfacing (4 hrs)

7.Internet of Things (IoT)and Embedded system (3hrs)

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1.Introduction to Embedded system

- ✓ **Introduction to Embedded System**
- ✓ **General Characteristics of Embedded System**
- ✓ **Classification of Embedded System**
- ✓ **Essential Components of Embedded System**
- ✓ **Overview of Processors and Hardware Units in an Embedded system**
- ✓ **Application of Embedded System**

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What is a system?

- A system is a way of working, organizing or doing one or many tasks according to a fixed plan, program or set of rules.
- For example: Watch is a time display system with parts: Hardware, Needles, Battery, Dial, Chassis and Strap.

Rules:

- ✓ All needles move clockwise direction.
- ✓ A thin needle rotates every second.
- ✓ A long needle rotates every minute.
- ✓ A short needle rotates every hour.
- ✓ All needles return to the original position after 12 hours.

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Introduction to Embedded System

- ✓David E. Simon “People use the term embedded system to mean any computer system hidden in any of these products.”

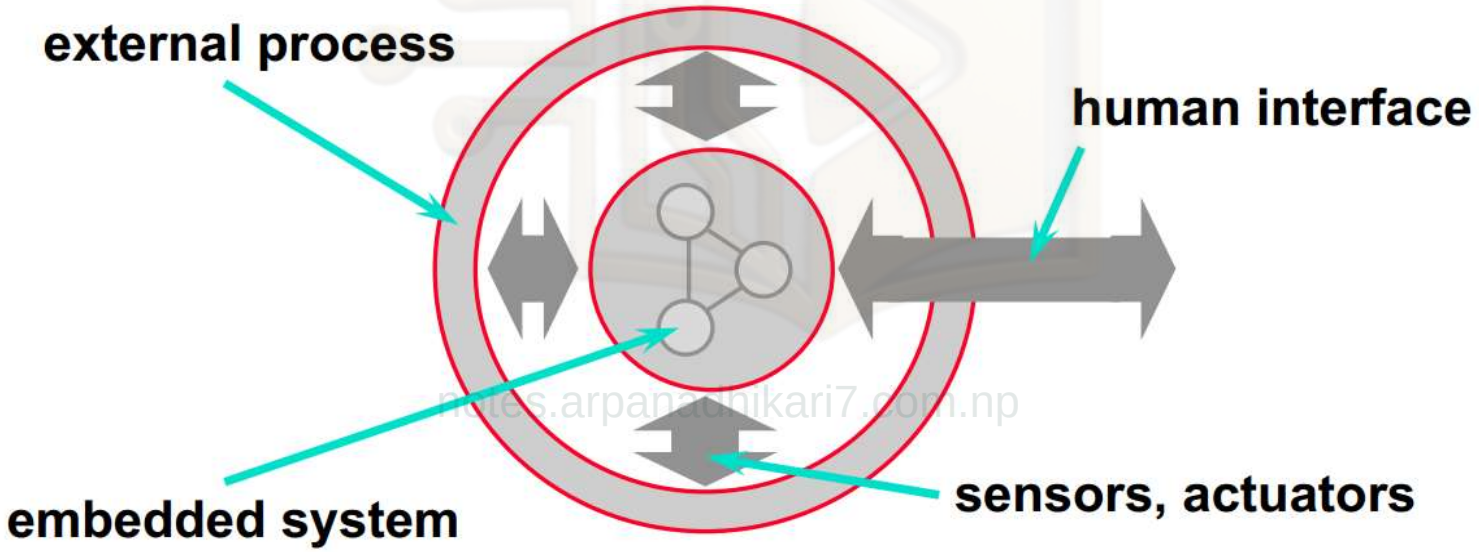
- ✓Todd D. Morton “Embedded Systems are electronic systems that contain a microprocessor or microcontroller, but we do not think of them as computers – the computer is hidden or embedded in the system.”

- ✓Tim Wilmshurst
  - ✓“An embedded system is a system whose principal function is not computational, but which is controlled by a computer embedded within it. The computer is likely to be a microprocessor or microcontroller. The word embedded implies that it lies inside the overall system, hidden from view, forming an integral part of greater whole. ”
  - ✓“An embedded system is a microcontroller-based, software-driven, reliable, real time control system, autonomous, or human- or network-interactive, operating on diverse physical variables and in diverse environments, and sold into a competitive and cost-conscious market”.

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Introduction to Embedded System

- ✓ **Any device that includes a computer but is not itself a general-purpose computer(Mainframe or Desktop).**
- ✓ **Respond to the external events.(e.g. someone pushes an elevator button.)**



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Introduction to Embedded System

- ✓ **Part of the large system.**
- ✓ **Expected to function without human intervention.**
- ✓ **Their work is subject to deadlines.**
- ✓ **Combination of software and hardware.**
- ✓ **Current ES are Multi-core, Multi-Platform mostly in real time and at Low Power Consumption.**

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Introduction to Embedded System

- An embedded system is a special-purpose system in which the computer is completely encapsulated by the device it controls.
- Unlike a general-purpose computer, such as a personal computer, an embedded system performs pre-defined tasks, usually with very specific requirements.
- Since the system is dedicated to a specific task, design engineers can optimize it, reducing the size and cost of the product.
- Embedded systems are often mass-produced, so the cost savings may be multiplied by millions of items
- Handheld computers or PDAs are generally considered embedded devices because of the nature of their hardware design, even though they are more expandable in software terms.

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Examples of embedded system

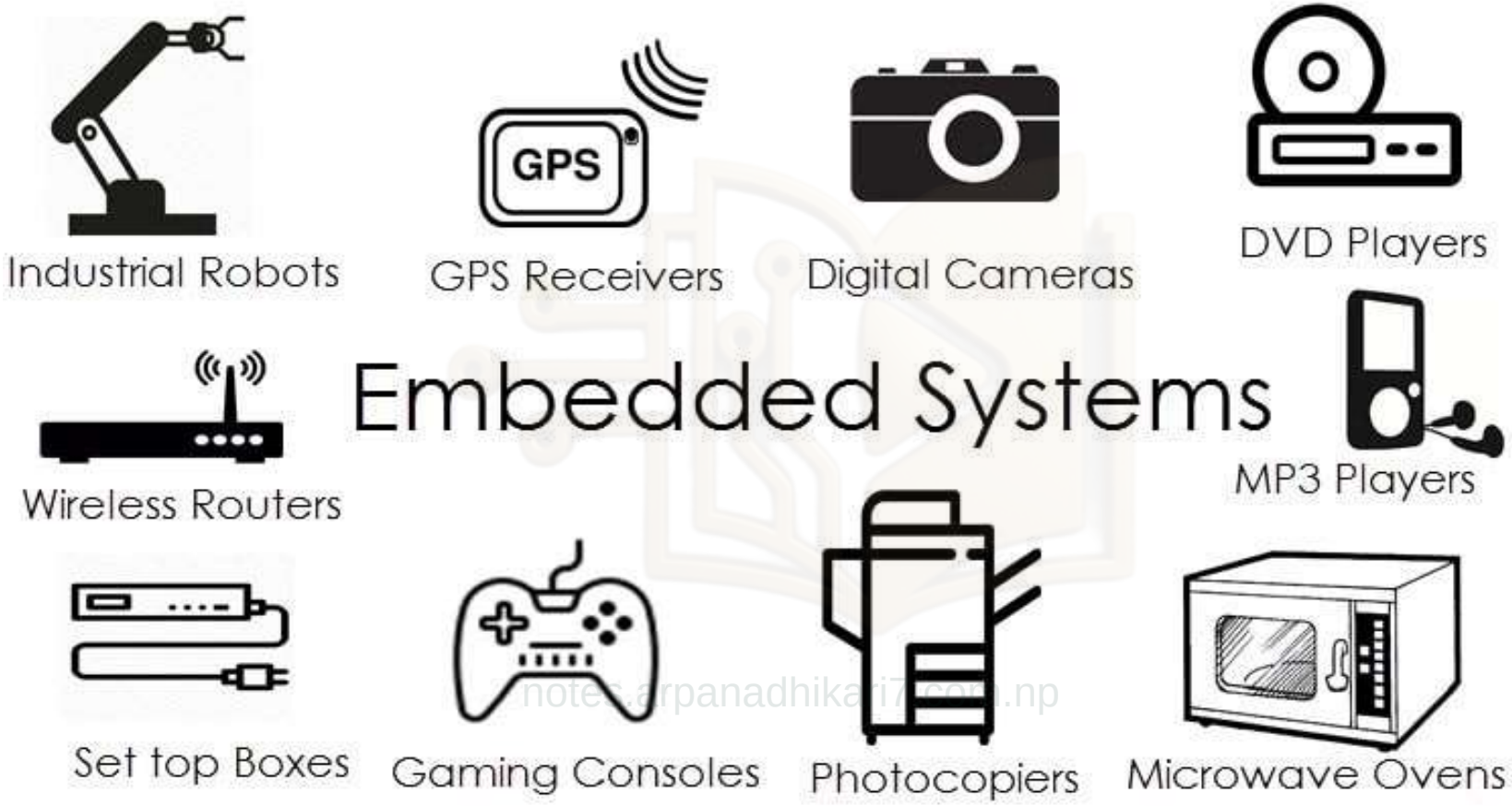
- Anti-lock brakes
- Auto-focus cameras
- Automatic teller machines
- Automatic toll systems
- Automatic transmission
- Avionic systems
- Battery chargers
- Camcorders
- Cell phones
- Cell-phone base stations
- Cordless phones
- Cruise control
- Curbside check-in systems
- Digital cameras
- Disk drives
- Electronic card readers
- Electronic instruments
- Electronic toys/games
- Factory control
- Fax machines
- Fingerprint identifiers
- Home security systems
- Life-support systems
- Medical testing systems

- Modems
- MPEG decoders
- Network cards
- Network switches/routers
- On-board navigation
- Pagers
- Photocopiers
- Point-of-sale systems
- Portable video games
- Printers
- Satellite phones
- Scanners
- Smart ovens/dishwashers
- Speech recognizers
- Stereo systems
- Teleconferencing systems
- Televisions
- Temperature controllers
- Theft tracking systems
- TV set-top boxes
- VCR's, DVD players
- Video game consoles
- Video phones
- Washers and dryers



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Examples of embedded system



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Characteristics of ES

The main characteristics of embedded system are:

- ✓ Single-functioned
  - ✓ Executes a single program, repeatedly
- ✓ Tightly-constrained
  - ✓ Low cost, low power, small, fast, etc.
- ✓ Real-time
  - ✓ Must compute certain results in real-time without delay.
  - ✓ In Hard Real time any delay in deadline could result in a catastrophe.

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Characteristics of ES

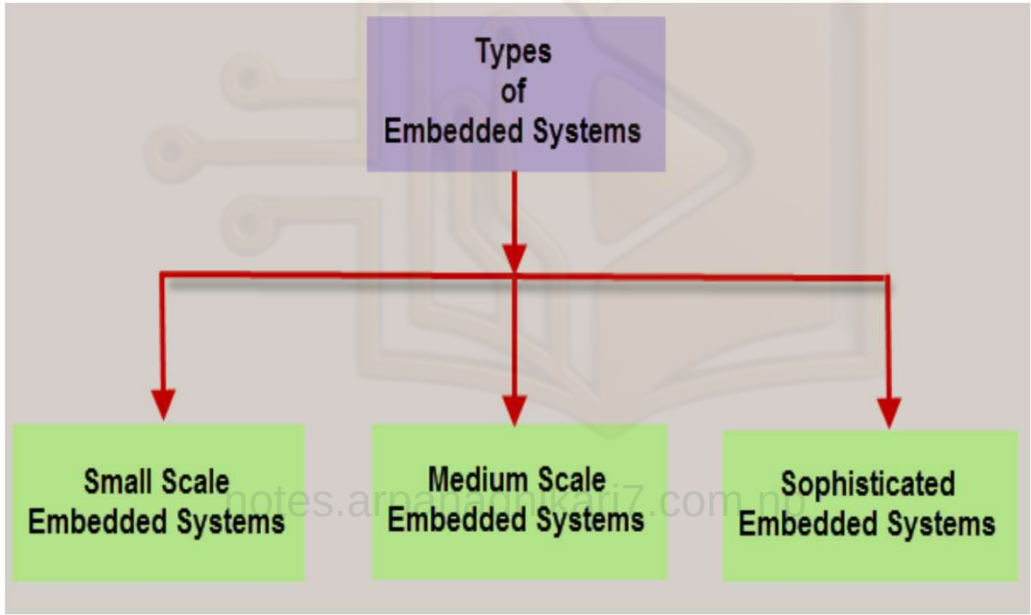
- ✓ Must be Dependable
  - ✓ Reliability, Maintainability, Availability, Safety, Security.
- ✓ Reactive
  - ✓ Continually reacts to changes in the system's environment.
- ✓ Sophisticated Functionality
  - ✓ Often have to run sophisticated or multiple algorithms for example cellphones, laser printer.

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Classification of ES

- Embedded systems are classified into three:



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Classification of ES

➢ Small Scale ES:

- ✓Single 8-bit or 16-bit Microcontroller.
- ✓Little Hardware and Software complexity.
- ✓They may be battery operated.
- ✓Usually C may be used to develop such system.
- ✓Programming Tools:
  - ✓ Editor, Assembler and Cross Assembler.
- ✓ Example: Stepper motor controller for a robotics system, Washing system, Computer Mouse, Printer, Scanner, Remote controller etc…

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Classification of ES

➢Medium Scale ES:

- ✓Single or few 16 or 32-bit microcontrollers or Digital Signal Processors(DSP) or Reduced Instruction Set Computers(RISC).
- ✓Both Hardware and Software Complexity.
- ✓Programming tools
  - ✓RTOS, Source Code Engineering tool, Simulator, Debugger and Integrated Development Environment(IDE).
- ✓Example: Banking systems like ATM and Credit card transactions, Entertainment systems like Video and Music system etc…

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Classification of ES

➢Sophisticated ES:

- ✓Enormous Hardware and Software complexity.
- ✓May need scalable processor or configurable processor and programming logic arrays.
- ✓Constrained by the processing speed available in their hardware units.
- ✓Example: Real time video processing system, Speech or Multimedia processing system.
- ✓Programming Tools
  - ✓For these systems may not be readily available at reasonable cost or may not be available at all. A compiler or retargetable compiler might have to be developed for this.

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Essential Components of ES

- ✓ Embeds hardware.
  - ✓Processors, Timers, Interrupt Controller, I/O devices, Memories, Ports etc.
- ✓Embeds main application software generally into flash or ROM and the application software performs concurrently the number of tasks.
- ✓Embeds a real time operating system, which supervises the application software tasks running on the hardware and organizes the accesses to the system resources according to priorities and timing constraints of tasks in the system.

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Key Components of ES

The key components of an embedded system are:

- **microcontroller/microprocessor** -brain
- **sensors** - input device,
- **actuators** - output devices
- **peripherals -** include memory, communication interfaces, and power management units.

- The microcontroller processes data from sensors and sends instructions to actuators to perform a specific task.

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Key Components of ES

➢Microcontroller/Microprocessor

- The central processing unit (CPU) that executes instructions.
- Contains memory (RAM and ROM) and I/O ports for communication.
- Handles all the logic and processing for the system's function.

➢Sensors

- Devices that convert physical phenomena (like temperature, pressure, light, or motion) into electrical signals.
- They provide input data to the system for processing.
- Examples include temperature sensors, accelerometers, and proximity sensors.

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Key Components of ES

➢Actuators

- Devices that convert electrical signals from the microcontroller into physical actions.
- They are the output components that control and manipulate physical processes.
- Examples include motors, solenoids, LEDs, and relays.

➢Peripherals

- These are a wide range of additional components that support the system's function, including:
- Memory: Stores the program and data (e.g., Flash, ROM, RAM).
- Input/Output (I/O) Ports: The microcontroller's interface to the outside world for communication.
- Communication Interfaces: Allow the system to communicate with other devices (e.g., UART, SPI, I2C).
- Timers and Counters: Track time, manage events, and generate delays.
- Power Management Units: Control and regulate the power supply to the system.

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Overview of Processors in ES

- ✓General Purpose Processor
  - ✓Microprocessor
    - ✓Intel : 8085, 8086, 80186, 80188, 80286,80386
    - ✓Motorola: 6800, 6809, G3, G4,G5
  - ✓Microcontroller
    - ✓Intel: 8032, 8051, 8052
    - ✓PIC: 8-bit PIC 16, PIC18, 16-bit DSPIC33/PIC24
    - ✓Motorola: 6811
  - ✓Embedded Processor
    - ✓AndeScore N9/10/12, ARM 7/9/11, Intel i960, AMD 29050.
  - ✓Digital Signal Processor
    - ✓PAC, TMS32oCXX, SHARC, Motorola 5600XX.

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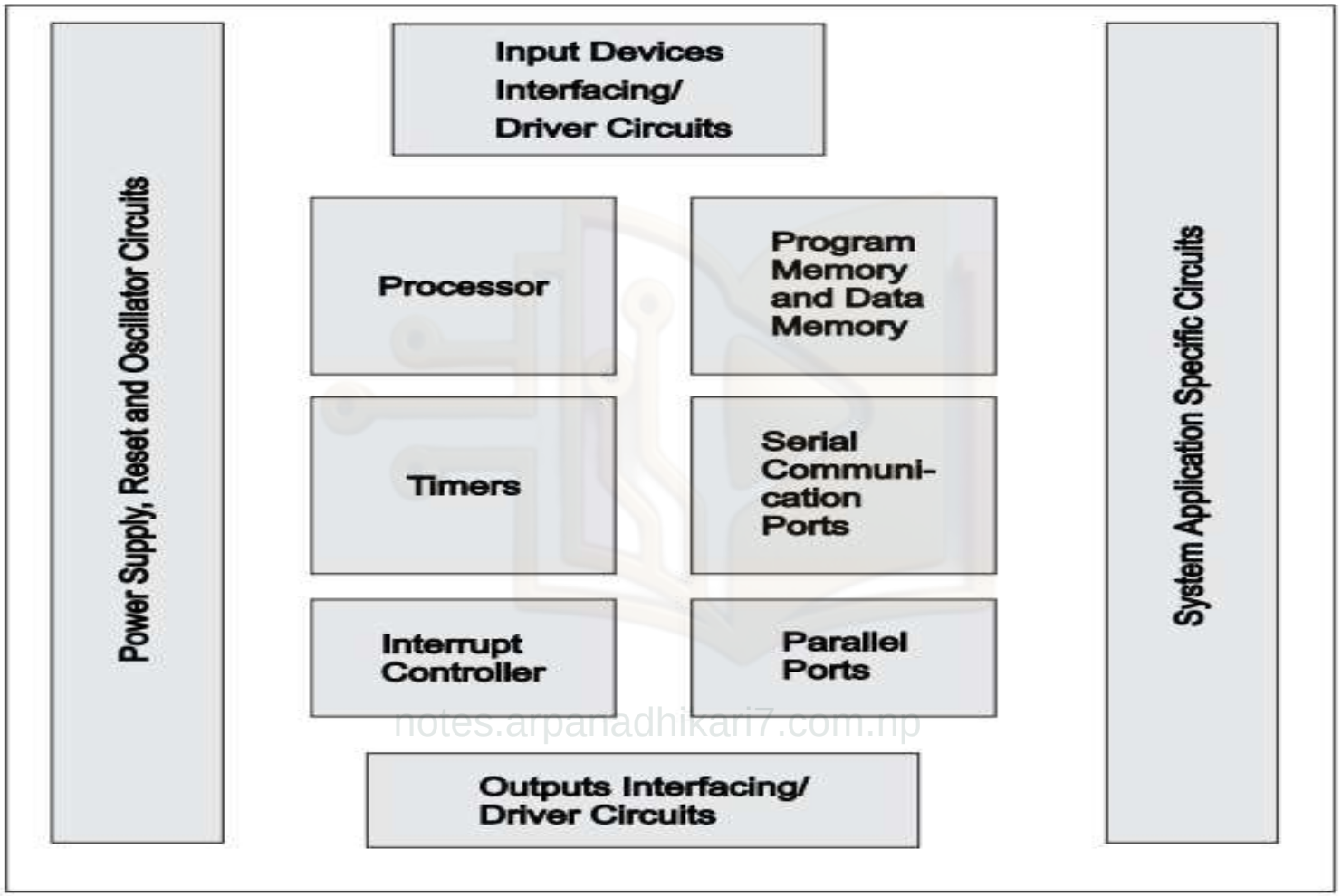
Overview of Processors in ES

- ✓Application-Specific Processor:
  - ✓Dedicated to specific tasks like image compression and provides faster solution.
  - ✓Used as an additional processing unit for running the application in place of using embedded software.
  - ✓Example: IIM7100, W3100A.

- ✓Single-Purpose Processor
  - ✓ Digital circuit designed to execute only a single program.
  - ✓Contains only the components to execute a single program.
  - ✓No program memory.
  - ✓ Example: coprocessor, accelerator or peripheral.

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Hardware Units in Embedded System

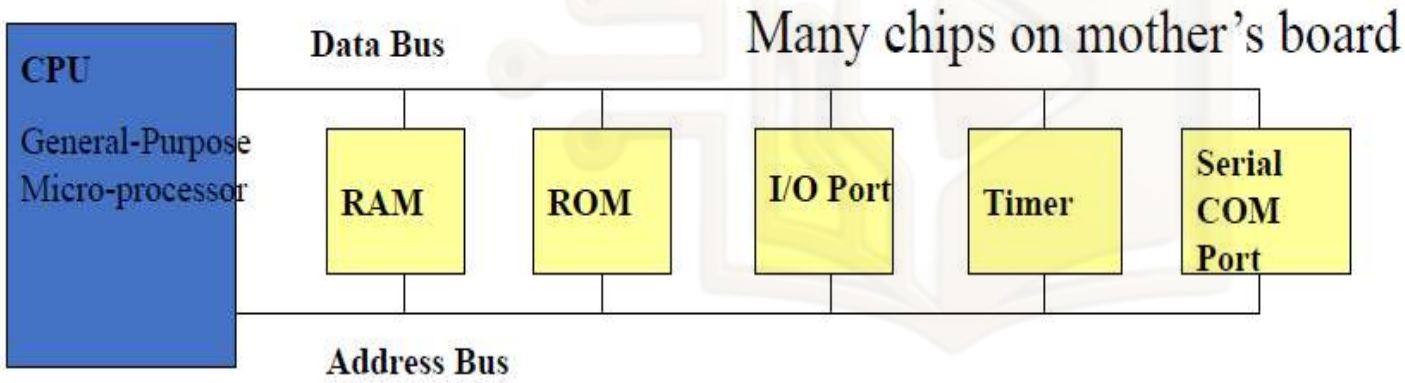


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General-purpose microprocessor

- CPU for Computers
- No RAM, ROM, I/O on CPU chip itself
- Example ： Intel’s x86, Motorola’s 680x0



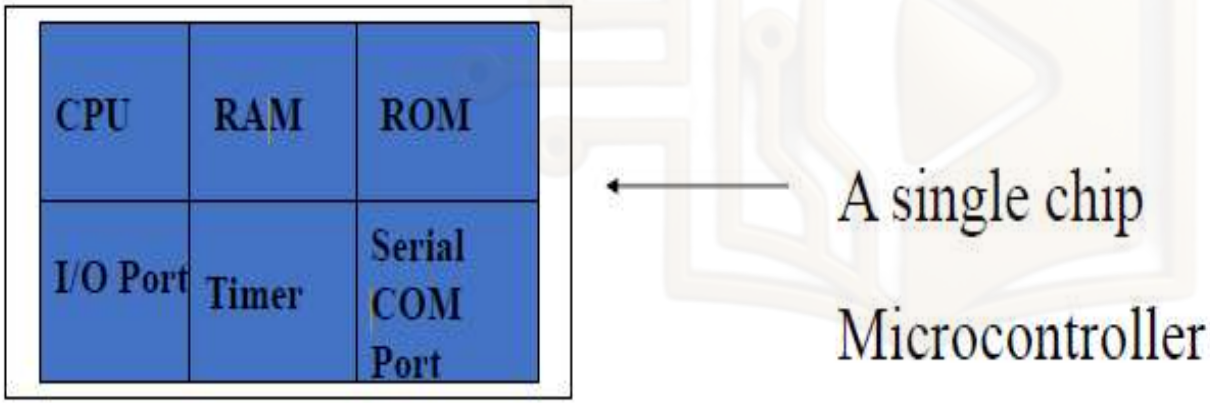
General-Purpose Microprocessor System

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Microcontroller

- A smaller computer
- On-chip RAM, ROM, I/O ports, timers etc
- Example ： Motorola’s 6811, Intel’s 8051, Zilog’s Z8 and PIC 16X



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Microprocessor vs Microcontroller

Microprocessor

- CPU is stand-alone, RAM, ROM, I/O, timer are separate
- designer can decide on the amount of ROM, RAM and I/O ports.
- expensive

Microcontroller

- CPU, RAM, ROM, I/O and timer are all on a single chip
- fix amount of on-chip ROM, RAM, I/O ports
- for applications in which cost, power and space are critical

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General Purpose Computing System Vs. Embedded System

General Purpose Computing System	Embedded System
A system which is a combination of a generic hardware and a General Purpose Operating System for executing a variety of application	A system which is a combination of special purpose hardware and embedded OS for executing a specific set of application
Contains a General Purpose Operating System (GPOS)	May or may not contain an operating system for functioning
Applications are alterable (programmable) by the user (It is possible for the end user to re-install the operating system, and also add or remove user application)	The firmware of the embedded system is pre- programmed and it is non- alterable by the end user (there may be exceptions for system supporting OS kernel image flashing through special hardware settings)
Performance is the key deciding factor in the selection of the system. Always, 'Faster is Better'	Application-specific requirement (like performance, power requirements, memory usage, etc.) are the key deciding factors
Response requirements are not time-critical	For certain category of ESs like mission critical systems, the response time requirement is highly critical
Need not be deterministic in execution behavior	Execution behavior is deterministic for certain types of ESs like 'Hard Real Time' systems

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Application of ES

- ✓Automotive Electronics
- ✓Aircraft Electronics
- ✓Trains
- ✓Telecommunication
- ✓Medical Systems
- ✓Military Applications
- ✓Authentication Systems
- ✓Consumer Electronics e.g MP3 Audio, digital Camera, home electronics etc.
- ✓Robotics etc…

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Purpose of Embedded Systems

- Data Collection/ Storage/ Representation
- Data Communication
- Data (signal) processing
- Monitoring
- Control
- Application specific user interface

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